

# AYUSHI MANDAL

Kolkata, West Bengal, India | ayushistacks@gmail.com | 9433874227 | [LinkedIn](#) | <https://portfolio-ayushi.vercel.app> | [GitHub](#)

## SUMMARY

B.Tech Information Technology undergraduate specializing in Software Engineering, Full-Stack Development, Android Development, Data Science, and Artificial Intelligence. Proficient in designing and developing end-to-end web and mobile applications using React, React Native, Next.js, Node.js, Kotlin, Jetpack Compose, TypeScript, Python, PostgreSQL, and modern software engineering practices. Leveraging Data Structures & Algorithms, Object-Oriented Programming, and Machine Learning to build efficient, scalable full-stack and mobile applications.

## EXPERIENCE

### Co-founder & Chief Product Officer, Haemologix

Jan 2025 - Present | Kolkata, India

- Designed and developed 5+ web and mobile applications across healthcare, emergency response, education, and e-commerce, supporting 200+ active users and reducing manual coordination time by 30% across deployed platforms.
- Built full-stack applications using React, Next.js, Node.js, Express, TypeScript, PostgreSQL, and MongoDB, implementing REST APIs, authentication, database-driven workflows, and responsive interfaces.
- Optimized application workflows and database interactions, achieving up to 40% improvement in response performance across project implementations through query optimization, API refinement, and efficient data handling.
- Managed end-to-end development across client and independent projects, delivering 90%+ of planned features within project timelines while handling requirements, architecture, testing, Git/GitHub workflows, debugging, and deployment.

## EDUCATION

### Bachelor of Technology in Information Technology

2028

University of Calcutta

Cultural Secretary | Member of the Design Department

### CBSE (Class XII)

2024

B.D.M. International

Earned Gold, Silver & Bronze medals in IMO, NSO & IEO for 3 consecutive years, with 2nd-level qualification, while actively participating in Interact, Cultural, Debate, Science & Sports Clubs

## SKILLS

Full-Stack Development, Android Development, Data Science, Artificial Intelligence

**Languages:** C, C++, Java, TypeScript, JavaScript, Python, Kotlin, SQL

**Frontend:** React, Next.js (App Router), Tailwind CSS, Jetpack Compose

**Mobile:** React Native, Expo, Android

**Backend & APIs:** Node.js, Express, Zod, REST/OpenAPI Design, Serverless Functions

**Databases & ORMs:** PostgreSQL (Neon), MongoDB, SQLite, Cloudflare D1, Prisma, Drizzle ORM, Room

**Auth & Security:** OAuth (Google/GitHub), NextAuth, bcrypt, Webhook Signature Verification

**Cloud & DevOps:** Vercel, Cloudflare Workers, Gradle, pnpm Workspaces, Git, GitHub

**Data & AI:** Pandas, NumPy, Matplotlib, OpenCV, Tensor flow, Scikit-learn, Seaborn, AI-Assisted Review Pipelines, LLM-Integrated Dashboards

## PROJECTS

### ResQ - Emergency Coordination Platform

Jan 2025-Mar 2025

- Built a full-stack emergency coordination platform connecting individuals, volunteers, and organizations during local disasters via a shared request/response system with role-aware workflows for four user types, processing 30+ simulated emergency requests during initial testing.
- Designed a deterministic matching engine ranking requests against available resources by category, distance, urgency, capacity, and age.
- Implemented secure authentication (NextAuth + bcrypt), protected API routes with ownership-based access control, and an interactive Leaflet map for geographic discovery, backed by a normalized 10-entity Prisma schema.

### CivicLens – Civic Issue Reporting App (Android)

Jun 2025-Sep 2025

- Built a native Android civic-issue reporting app with a local-first architecture, letting residents report, verify, and track infrastructure issues on a map-first interface.

- Designed a guided reporting flow (Capture, AI-assisted detection review, Location, Submit) with a gamified civic-points, achievements, and leaderboard system to drive engagement, achieving 85% detection accuracy for common infrastructure issue types during testing.
- Implemented local persistence with Room across users, issues, verifications, and resolutions, paired with a TypeScript backend workspace exposing OpenAPI-spec'd, Zod-validated endpoints and a generated React client.

#### NOVI – AI Personal Intelligence Environment

Jan 2026-Mar 2026

- Built a unified "Life Canvas" dashboard that syncs and normalizes data from Gmail, Calendar, Drive, and GitHub (repos, issues, PRs, commits) into a single source-backed intelligence view, indexing 500+ data points across connected accounts.
- Implemented secure OAuth integrations for Google and GitHub with encrypted credential storage, webhook signature verification, and strict separation between demo and connected-account data.
- Designed an "Ask Novi" command surface combining local matching with source-backed connected search, backed by a normalized entity/relationship schema on Cloudflare D1.

#### Candidate OS – Exam Application & Candidate Management Simulation Platform

Jul 2026-Sep 2026

- Architected a full-stack simulation of a national exam administration system, handling application intake, admit card generation, results processing, and multi-center exam logistics.
- Built an in-process background job queue for asynchronous workflows under serverless constraints, plus hand-rolled SHA-256 hashing for consistent data integrity across server, edge, and browser environments.
- Designed an audit-trail and PII-scanning system with hash-matched document previews, and load-tested the platform simulating 5,000 concurrent applications in ~2.5 seconds to validate scalability.